Standing Long Jump Assessment Based on Human Pose Estimation and Gradient Boosting Regression Trees

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Abstract. This study proposes an intelligent standing long jump assessment framework that integrates human pose estimation, anthropometric and body-composition characteristics, and machine learning. Thirty-three human keypoints were smoothed using a five-frame centered moving average. Take-off and landing were identified by jointly considering foot clearance, displacement of a proxy center of mass, and velocity changes. Exploratory Pearson correlation analysis was conducted on 32 repeated observations from eight participants, and jump distance was predicted using a gradient boosting regression tree evaluated by leave-one-participant-out cross-validation. Basal metabolic rate, body mass, and skeletal-muscle-related variables showed strong positive associations with jump distance. The model achieved a root mean square error of 0.129 m, a mean absolute error of 0.103 m, and an R^2 of 0.803. For a 6-year-old target participant, the predicted distance was 1.250 m. Under scenario-based target technical parameters, the simulated distance increased to 1.396 m. The proposed framework links movement-event identification, performance prediction, and training recommendations, but its generalizability remains limited by the small sample, repeated-measures structure, uncalibrated image coordinates, and lack of prospective intervention validation.

ACM CCS (2012) Classification:

1. Computing methodologies → Computer vision → Activity recognition and understanding

2. Computing methodologies → Machine learning → Machine learning algorithms

3. Applied computing → Education → Computer-assisted instruction

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1. Introduction

The standing long jump is a basic item in the National Student Physical Fitness and Health Standard and provides an integrated indication of lower-limb explosive power, whole-body coordination, trunk control, and landing stability. Traditional assessment relies primarily on the final jump distance and teachers' on-site observation. Although this approach is convenient, judgments about brief movement details during take-off, flight, and landing are partly subjective, and continuous quantitative data on individual performance are difficult to obtain. In recent years, computer vision and human pose estimation have increasingly been applied to intelligent student fitness testing and movement analysis. Human keypoint coordinates can be acquired using ordinary cameras, providing technical support for non-contact quantitative assessment throughout the standing long jump [1-4]. Machine learning can also integrate body-composition, movement, and performance data to support data-driven prediction and the identification of variables associated with performance [5-7]. Existing studies tend to emphasize final distance measurement, provide insufficient kinematic interpretation of the complete movement, separate body-composition analysis from technique analysis, and offer limited linkage between assessment results and training guidance. To address these limitations, this study constructed an analytical framework that integrates pose-derived kinematics, anthropometric and body-composition characteristics, and gradient boosting regression. The framework supports take-off and landing identification, exploratory association analysis, jump-distance prediction, and scenario-based training recommendations, thereby providing an interpretable technical approach for intelligent standing long jump assessment in schools.

2. Methodology

2.1. Dataset and Participants

This study conducted a secondary analysis of anonymized human-keypoint data, anthropometric and body-composition measurements, and standing long jump records supplied with the task. The dataset comprised 11 participants. Participants 1 and 2 were used to examine the take-off and landing identification procedure and to compare movement characteristics during the flight phase. Participants 3-10 comprised eight participants with four records each, yielding 32 repeated observations that included pre- and post-training movement and performance measurements; these records were used for exploratory association analysis and construction of the jump-distance prediction model. Participant 11 was a 6-year-old target participant for prediction, for whom only human-keypoint trajectories and individual body-composition information were available. The correspondence between frame number and time indicated a video frame rate of 30 frames/s.

2.2. Pose-Coordinate Preprocessing and Kinematic Feature Extraction

Coordinates produced by human pose estimation exhibit frame-to-frame jitter, particularly during rapid movements such as take-off and landing. A five-frame centered moving average was therefore applied to smooth the horizontal and vertical coordinates of keypoint j at frame t [8,9]:

$$\tilde{x}_{j,t} = \frac{1}{5}\sum x_{j,t+k}, \tilde{y}_{j,t} = \frac{1}{5}\sum y_{j,t+k} \tag{1}$$

where the tildes denote the smoothed horizontal and vertical coordinates of keypoint j at frame t. At the sequence boundaries, the mean was calculated using the frames actually available within the window. Foot analysis used the bilateral ankle, heel, and toe keypoints (keypoints 27-32), with heel and toe keypoints 29-32 used primarily for take-off and landing identification. The locations and numbering of the 33 human keypoints are shown in Figure 1. A proxy center of mass (pCoM) for the trunk was defined as the mean position of the left and right shoulders (keypoints 11 and 12) and the left and right hips (keypoints 23 and 24):

$$C_t = \frac{1}{4}\sum P_{i,t}, i \in \{11,12,23,24\} \tag{2}$$

where $P_{i,t}$ is the two-dimensional coordinate vector of keypoint i at frame t. Horizontal velocity v_x and vertical velocity v_γ were calculated from pCoM coordinates in adjacent frames. The extracted features included horizontal displacement, vertical pCoM change, foot-clearance height, take-off velocity, take-off angle, knee angle, ankle angle, trunk forward-lean angle, and arm-swing amplitude. Because the original videos were not camera-calibrated, displacement and velocity were expressed in pixels and pixels/s, respectively, whereas angular variables were expressed in degrees.

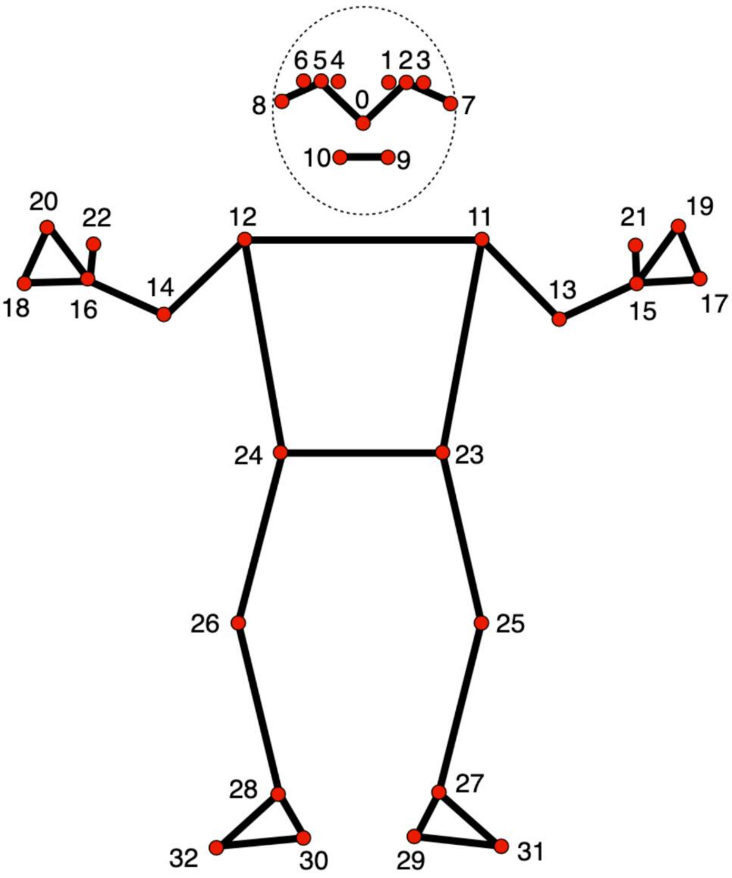

Figure 1. Numbering and Configuration of the 33 Human Keypoints.

2.3. Take-Off and Landing Identification and Flight-Phase Segmentation

Take-off is generally accompanied by increased foot clearance, increased horizontal pCoM velocity, and a rapid change in the vertical pCoM coordinate. At landing, foot height returns to the ground baseline and vertical velocity changes markedly. To reduce the influence of any single threshold, horizontal displacement, vertical displacement, foot-clearance height, and velocity features were first directionally aligned and standardized and were then combined into a multi-index composite score. Within the candidate movement window, the frame corresponding to the peak composite score was identified as the take-off frame. The first post-take-off frame that simultaneously satisfied the conditions of foot height returning to baseline, termination of the pCoM descent trend, and an abrupt velocity change was identified as the landing frame.

Let t_0 denote the take-off frame and t_1 the landing frame. The flight time T was calculated as follows:

$$T = \frac{t_1 - t_0}{30} \tag{3}$$

For standardized comparison across participants, normalized flight time was descriptively divided into an approximate ascent phase [0, 0.3T], an apex phase [0.3T, 0.7T], and an approximate descent phase [0.7T, T]. Mean or extreme values of vertical pCoM change, knee angle, and trunk forward-lean angle were calculated within each phase. This segmentation was used for descriptive comparison and should not be interpreted as strict kinematic phase division based on the zero crossing of vertical velocity.

2.4. Construction of Body-Composition Indices and Exploratory Association Analysis

The anthropometric and body-composition data included height, body mass, body-fat percentage, muscle mass, skeletal muscle mass, muscle percentage, basal metabolic rate, nutritional status, and body type. Missing numerical values were imputed using the median. Outliers were identified using the interquartile-range rule and winsorized to the corresponding boundary. Sex, nutritional status, and body type were encoded as binary or ordinal categorical variables. To improve interpretation beyond individual raw measurements, the following composite indices were constructed:

$$\text{Body mass index (BMI)} = \text{body mass (kg)} / \text{height}^2\ (\text{m}^2)$$

$$\text{Skeletal muscle index} = \text{skeletal muscle mass (kg)} / \text{height}^2\ (\text{m}^2)$$

$$\text{Relative muscle mass} = \text{muscle mass (kg)} / \text{body mass (kg)}$$

$$\text{Explosive-power potential index} = \text{muscle percentage} \times (100 - \text{body-fat percentage}) / 100$$

Pearson correlation coefficients were used to examine linear associations between anthropometric, body-composition, and technical variables and jump distance. The percentage change in jump distance before and after training was used to describe individual improvement following technical correction. The 32 records were treated as observational units for correlation analysis; however, repeated observations from the same participant were not independent. Because only eight participants were available for modeling, no significance tests based on large-sample assumptions were performed. The correlation coefficients were used only for exploratory ranking and description of variable relationships and should not be interpreted as causal effects.

2.5. Jump-Distance Prediction Model

Standing long jump prediction was formulated as a supervised regression task. Input features comprised anthropometric and body-composition variables, kinematic variables from take-off and the flight phase, and technique-related angular variables. Continuous variables were standardized using Z scores. In each cross-validation fold, the mean and standard deviation were estimated from the training data only and then applied to the corresponding validation data. A gradient boosting regression tree (GBRT) was used as the predictor, with 100 weak learners, a maximum tree depth of 3, a learning rate of 0.1, and a random seed of 42.

Given the limited sample, leave-one-participant-out cross-validation (LOPO-CV) was adopted. In each iteration, all records from one participant were used as the validation set, and all records from the remaining seven participants were used as the training set. The procedure was repeated eight times to obtain out-of-fold predictions while preventing records from the same participant from appearing in both the training and validation sets. Model performance was evaluated using the root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2). The interval estimate for Participant 11 was constructed from the dispersion of the cross-validation residuals and is therefore termed an approximate 95% prediction interval rather than a parametric confidence interval.

2.6. Training Recommendation Generation and Scenario Simulation

To generate training recommendations, modifiable technical variables were first selected on the basis of the exploratory associations and movement-difference analysis. Gradual target values for Participant 11 were then specified with reference to the range of pre- to post-training technical changes observed in the modeling participants. In the scenario simulation, anthropometric and body-composition variables were held constant, while the selected technical variables were replaced by their target values and entered into the model to obtain the predicted jump distance under the target-parameter scenario. The 18-week program translated the model-based recommendations into phased instructional tasks; it was not used in model training and did not establish a causal relationship between training duration and performance improvement.

3. Results

3.1. Take-Off and Landing Identification and Flight-Phase Differences

The take-off and landing frames, flight times, and measured jump distances for Participants 1 and 2 are summarized in Table 1. Their flight times were similar; however, the horizontal and vertical take-off velocities of Participant 1 were 106.2 pixels/s and 75.2 pixels/s, respectively, which were substantially higher than the corresponding values of 45.7 pixels/s and 28.1 pixels/s for Participant 2. These findings indicate that flight time alone cannot explain final jump distance; the take-off velocity profile and flight-phase posture should also be included in the assessment.

Table 1. Take-Off, Landing, Flight-Time, and Performance Parameters for Participants 1 and 2.

Participant	Take-off	Landing	Flight time (s)	Jump distance (m)
Participant 1	Frame 122 (4.067 s)	Frame 148 (4.933 s)	0.867	1.58
Participant 2	Frame 160 (5.333 s)	Frame 188 (6.267 s)	0.933	1.15

Phase-specific comparison showed that during the approximate ascent phase, Participant 1 had a knee angle of 127.3° and a trunk forward-lean angle of 25.2°, whereas the corresponding values for Participant 2 were 116.0° and 49.3°. The greater forward lean of Participant 2 after take-off suggests differences in trunk control and velocity direction. During the approximate descent phase, the knee angle of Participant 1 decreased to 39.8°, indicating more pronounced knee-flexion cushioning, whereas the corresponding value for Participant 2 was 58.4°, suggesting less adequate landing cushioning and forward-reaching preparation. Combining the pCoM trajectory with joint-angle information helps explain why similar flight times can be associated with different jump distances.

3.2. Exploratory Associations of Body-Composition and Technical Variables

The leading associated variables and their correlation coefficients are presented in Table 2. Basal metabolic rate, body mass, skeletal muscle mass, skeletal muscle index, relative muscle mass, and muscle percentage all had correlation coefficients greater than 0.73 with jump distance. Basal metabolic rate showed the highest coefficient ($r = 0.860$), followed by body mass ($r = 0.813$) and skeletal muscle mass ($r = 0.797$). Among the technical variables, post-training minimum ankle angle, take-off velocity, and minimum trunk forward-lean angle showed moderate positive correlations with jump distance, whereas arm-swing amplitude showed a negative correlation. Because the data included repeated observations from the same participants, these coefficients reflect a mixture of between-participant differences and within-participant changes.

Table 2. Variables Most Strongly Associated with Standing Long Jump Distance (Exploratory Analysis).

Variable	Correlation coefficient, r	Category
Basal metabolic rate (kcal)	0.860	Body composition
Body mass (kg)	0.813	Body composition
Skeletal muscle mass (kg)	0.797	Body composition
Skeletal muscle index	0.790	Body composition
Relative muscle mass	0.739	Body composition
Muscle percentage (%)	0.738	Body composition
Post-training minimum ankle angle	0.695	Technique
Post-training take-off velocity	0.669	Technique
Post-training minimum trunk forward-lean angle	0.666	Technique
Post-training arm-swing amplitude	-0.750	Technique

Most participants improved after training, although the magnitude of improvement varied considerably. Participants with lower initial performance generally had greater scope for technical correction, whereas further improvement among higher-performing participants depended more heavily on body-composition foundations and fine technical adjustment. The negative correlation for arm-swing amplitude may reflect movement compensation, differences in body scale, or a small number of atypical participants; it does not imply that arm swing should be mechanically reduced. Training should instead consider the timing and direction of arm swing and its coordination with lower-limb extension.

3.3. Model Performance and Prediction for Participant 11

As shown in Table 3, LOPO-CV produced an RMSE of 0.129 m, an MAE of 0.103 m, and an R^2 of 0.803 for the GBRT model. Most participant-level prediction errors were within ±0.15 m. The larger deviation observed for Participant 6 suggests that unusual movement patterns or body-composition profiles may lie outside the effective learning range of a small-sample model. These values represent internal cross-validation performance in the current sample and do not replace external validation using an independent dataset.

Table 3. Leave-One-Participant-Out Cross-Validation Performance of the GBRT Model.

Metric	Value
RMSE (m)	0.129
MAE (m)	0.103
R^2	0.803

For Participant 11, take-off occurred at frame 176 (5.87 s) and landing at frame 188 (6.27 s), resulting in a flight time of 0.400 s and a take-off angle of approximately 32.1°. The pCoM trajectory was approximately parabolic, but the flight time was substantially shorter than those of Participants 1 and 2, suggesting that take-off velocity and flight-phase posture control were potential priorities for improvement. The model predicted a standing long jump distance of

1.250 m, with an approximate 95% prediction interval of 0.873-1.627 m. The wide interval reflects uncertainty arising from the limited training sample and differences in age and developmental status.

3.4. Training Recommendations and Scenario-Simulation Results

The target technical parameters and predicted percentage changes for Participant 11 are shown in Table 4. Flight time was increased from 0.400 s to 0.583 s, take-off angle was adjusted from 32.1° to 42.0°, arm-swing amplitude from 62.0° to 86.7°, take-off knee angle from 108.0° to 135.0°, and trunk forward-lean angle from 14.0° to 22.7°. These values represent reference states within the model scenario and should be approached gradually under teacher supervision rather than treated as mandatory thresholds for a single training session.

Table 4. Scenario-Based Target Technical Parameters for Participant 11.

Technical parameter	Initial value	Target value	Predicted change
Flight time (s)	0.400	0.583	45.8%
Take-off angle (°)	32.1	42.0	30.9%
Arm-swing amplitude (°)	62.0	86.7	39.8%
Take-off knee angle (°)	108.0	135.0	25.0%
Trunk forward-lean angle (°)	14.0	22.7	61.9%

The proposed 18-week program comprised three phases. Weeks 1-4 constituted the fundamental movement-establishment phase, with four sessions per week of 20-25 min each, focusing on movement decomposition, rhythm-based jumping drills, and distance-perception games. Weeks 5-10 constituted the technique-reinforcement phase, with three sessions per week of 25-30 min each, including integrated whole-movement practice, target-zone jumps, and comparison with peer demonstrations. Weeks 11-18 constituted the movement-consolidation phase, with three sessions per week of 30-35 min each, using simulated tests, personal-best challenges, and positive feedback to improve movement consistency. For children, movement quality and sustained interest should take priority; excessive loading, repeated high-intensity jumping, and repeated testing under fatigue should be avoided.

After the target technical parameters in Table 4 were entered into the model, the scenario-simulated jump distance increased from 1.250 m to 1.396 m, corresponding to an absolute increase of 0.146 m and a relative increase of 11.7%. This value is a model output under the target-parameter scenario rather than an observed outcome after 18 weeks of training, and it does not imply that the same improvement will necessarily be achieved in practice.

4. Discussion

4.1. Multi-Index Fusion Supports Robust Movement-Event Identification

Take-off and landing were identified using a combination of pCoM displacement, velocity changes, and foot-clearance features rather than foot height or a single keypoint alone. Multi-index fusion can reduce the risk of misclassification caused by short-term pose jitter, foot occlusion, or isolated coordinate anomalies. The identified events can also support analysis of knee angle, trunk angle, and pCoM trajectory, extending assessment from whether and when the participant jumped to how the movement was performed. Results for Participants 1 and 2 showed that although Participant 2 had a slightly longer flight time, both horizontal and vertical take-off velocities were lower and landing preparation was less adequate, resulting in a shorter jump distance. Standing long jump assessment should therefore consider take-off velocity, pCoM trajectory, trunk control, and landing technique simultaneously.

4.2. Body-Composition Foundations and Technical Variables Require Distinct Interpretation

Basal metabolic rate, skeletal muscle mass, and skeletal muscle index were strongly associated with jump distance, suggesting that overall development and muscular foundations may be important correlates of standing long jump performance. The positive association between body mass and jump distance should not be interpreted to mean that increasing body mass will improve performance; the relationship may be jointly influenced by age, height, muscle mass, and overall developmental status. In contrast, take-off velocity, knee angle, and trunk forward-lean angle are more modifiable and are therefore more appropriate targets for instructional feedback. The negative association between arm-swing amplitude and jump distance also requires caution. Arm swing contributes to momentum transfer and balance control, and the observed direction may have been influenced by swing timing, swing direction, body scale, and upper- and lower-limb coordination.

4.3. Prediction and Scenario Simulation Can Support Instruction

Gradient boosting regression trees can represent nonlinear relationships between anthropometric/body-composition variables and movement-technique variables. In the current small sample, the model demonstrated useful predictive capacity, with an RMSE of 0.129 m, an MAE of 0.103 m, and an R^2 of 0.803. Nevertheless, the results may be sensitive to individual observations and hyperparameter settings and should not replace direct physical-fitness measurement. The predicted jump distance of 1.250 m and prediction interval of 0.873-1.627

m for Participant 11 can support preliminary estimation of performance level and potential technical weaknesses. The resulting 18-week recommendation emphasizes take-off extension, coordination of arm swing, flight-phase posture, and landing cushioning. The value of 1.396 m must be understood only as a scenario-simulated result under the target technical parameters. Actual instruction should be adjusted dynamically using direct measurement, teacher observation, the child's age and developmental characteristics, and training-safety considerations.

4.4. Limitations and Directions for Further Research

This study has four principal limitations. First, only eight participants were used for modeling, and the 32 records had a repeated-measures structure; consequently, the stability of the correlations and performance metrics is limited. Second, take-off and landing identification was not compared with frame-by-frame manual annotation, so event-detection error could not be quantified. Third, the camera was not calibrated, which limits the comparability of pixel displacement and velocity across recording scales. Fourth, Participant 11 may have differed from the modeling participants in age and developmental status, and both the target parameters and performance gain were generated through scenario simulation rather than a prospective training intervention. Future studies should collect larger stratified samples, standardize recording conditions or implement scale calibration, include manual event annotations and an independent test set, and evaluate the effectiveness and safety of the training recommendations through actual interventions.

5. Conclusion

Using 33-point human pose coordinates, a five-frame centered moving average, and multi-index fusion, this study identified take-off and landing during the standing long jump and quantified movement and posture characteristics during the flight phase. Under the current small-sample and repeated-measures conditions, exploratory analysis showed strong or moderate associations of jump distance with basal metabolic rate, skeletal-muscle-related variables, and take-off velocity; these findings do not establish causality. LOPO-CV of the GBRT model yielded an RMSE of 0.129 m, an MAE of 0.103 m, and an R^2 of 0.803, and the predicted jump distance for Participant 11 was 1.250 m. Under the target technical parameters, the scenario-simulated distance was 1.396 m, and an 18-week phased training recommendation was developed. Human pose estimation and machine learning can therefore provide auxiliary support for standing long jump assessment and instructional feedback; however, event-identification accuracy, cross-setting comparability, model generalizability, and training effectiveness require further validation through larger samples, camera calibration, manual annotation, and actual intervention studies.

References

[1] Li, J. (2023). Technology-enabled intelligent physical fitness testing systems in college student physical fitness assessment. In Proceedings of the Fourth Academic Conference on Scientific Exercise for National Fitness and the International Forum on Exercise and Health (p. 536). (in Chinese)

[2] Gu, X., Wang, C. W., Xu, H., et al. (2023). A computer-vision-based intelligent physical fitness testing system for students. Information Technology and Informatization, (10), 186–193. (in Chinese)

[3] Geng, X. Y. (2023). Research and design of a deep-learning-based system for assessing long-jump movement standardization [Master's thesis, Shandong University of Technology]. (in Chinese)

[4] Li, S., Song, K. E., Ouyang, B. Q., et al. (2022). An intelligent assessment system for the standing long jump based on human pose recognition. Artificial Intelligence, (2), 75–87. (in Chinese)

[5] Shen, L. (2025). Research progress in methods for predicting the competitive performance of elite athletes. Chinese Sports Coach, 33(2), 15–18, 25. (in Chinese)

[6] Li, Y. J., & Ding, Q. (2025). Prediction of student performance based on machine learning. Logistics Technology, 48(7), 178–180. (in Chinese)

[7] Guo, C., Hong, Y. H., Liang, S. R., et al. (2025). Computer-vision-based intelligent physical fitness testing and auxiliary training. China Computer & Communication, 37(3), 161–163. (in Chinese)

[8] Liu, Y., Shen, S. N., Sun, Y., et al. (2020). A smoothing algorithm for elevation data. Science & Technology Vision, (11), 54–56. (in Chinese)

[9] Yang, H. X., Li, H., & Cao, J. (2018). Monitoring-data smoothing based on spline functions and its MATLAB implementation. Journal of Changchun Institute of Technology (Natural Sciences Edition), 19(3), 92–97, 120. (in Chinese)